Akshaj Raut

Boston, MA | raut.aks@northeastern.edu | LinkedIn | GitHub | Portfolio

Education

Northeastern University, Boston, MA April 2026
Master of Science in Robotics, Concentration: Electrical and Computer Engineering *GPA: 3.6/4*

- **Coursework:** Autonomous Field Robotics, Mobile Robotics, Computer Vision, Robot Sensing and Navigation

University of Mumbai, Mumbai, India May 2023
Bachelor of Engineering in Electronics and Telecommunication *GPA: 8.8/10*

- **Coursework:** Autonomous Vehicles, IoT and Industry 4.0, Wireless Networks

Skills

Robotics and Perception: SLAM (Visual / LiDAR / Visual-Inertial), Imitation Learning (ACT, Diffusion Policy), 3D Reconstruction, Gaussian Splatting, cuVSLAM (Isaac ROS), ROS2, Gazebo, RViz2, Isaac Sim, ArduPilot, GTSAM, OpenCV
Programming: Python, C++, Embedded C, PyTorch, Java, Dart
Hardware and Sensors: NVIDIA Jetson Orin Nano, Pixhawk, Raspberry Pi, Arduino, ESP32/8266, IMU, GPS, LiDAR, RGB/Depth Cameras
Tools: SolidWorks (CSWA, CSWP), KiCad, EAGLE, MATLAB, LabVIEW, Git/GitHub/GitLab

Work Experience

PicoStone Technologies September 2022 – December 2022
Hardware Development Intern *Mumbai, India*

- Engineered firmware for smart home lights using ESP-IDF optimised for stable WiFi connectivity and sub 200 ms response
- Designed a KiCad PCB for a WiFi LED controller, reducing board area by 20% and lowering cost by 10%
- Led testing with a 4-member hardware team to integrate smart lighting solutions, enhancing product reliability

Projects

LeRobot SO-101 Imitation Learning (ACT / Diffusion Policy) May 2026 – Present

- Brought up an SO-101 leader-follower teleoperation rig (LeRobot, HuggingFace) on real robotic hardware for pick-and-place demonstration collection
- Built a synchronized data pipeline capturing top-camera RGB and joint-state recordings across demonstrations for imitation learning
- Training Action Chunking Transformer (ACT) and Diffusion Policy models in PyTorch on the collected data and benchmarking autonomous pick-and-place success rate on the physical arm

Gaussian Splatting and LiDAR based 3D Scene Reconstruction January 2026 – March 2026

- Captured a 1658 image low texture indoor dataset and analyzed failure modes of COLMAP feature matching in featureless environments, leading to unstable camera pose estimation in Nerfstudio Gaussian Splatting pipelines
- Replaced failed SfM pose recovery with ARKit camera poses and LiDAR depth exported from Polycam, leveraging IMU and visual inertial tracking for robust pose estimation independent of image features
- Reconstructed a 2 cm voxel resolution corridor mesh (795K vertices) via Open3D ScalableTSDFVolume TSDF fusion, integrating depth maps and camera extrinsics to capture walls, floor, and ceiling in a featureless environment

Autonomous AI Drone May 2025 – August 2025

- Built a custom UAV on a Pixhawk flight controller running ArduPilot (ArduCopter 4.6.0), paired with an NVIDIA Jetson Orin Nano for onboard compute over 915 MHz telemetry and MAVLink; tuned the EK3 estimator for GPS, IMU, and barometer sensor fusion with autonomous waypoint navigation
- Integrated TF-Luna TOF LiDAR and RGB camera for object detection and obstacle avoidance; prototyped person-following and object lock-scan functionality to capture multi-angle images for 3D reconstruction and anomaly detection
- Currently implementing cuVSLAM (NVIDIA Isaac ROS) on the Jetson Orin Nano for real-time onboard visual SLAM and mapping

Live Firearm Detection and Alerting System using YOLOv8 March 2025 – April 2025

- Fine-tuned a YOLOv8n model on a custom 2376 image dataset, achieving 91% mAP@0.5, 100% precision, and 92% recall for firearm detection
- Deployed real-time inference on live webcam feeds at 30+ FPS with intelligent frame capture (5 pre/post-detection frames) for forensic analysis
- Architected an automated multi-channel alerting pipeline with Twilio API for instant SMS and voice alerts, reducing incident response time by 40%

ORB-SLAM3 on Autonomous Car November 2024 – December 2024

- Deployed ORB-SLAM3 stereo-inertial SLAM on the Nuance autonomous car, processing 20,000+ stereo frames from a multi-camera setup
- Collected, calibrated, and synchronized real-world data from stereo cameras and IMU for robust feature detection
- Achieved a consistent 5 mile trajectory with successful loop closure, maintaining <2% overall trajectory error